

LAND USE LAND COVER CLASSIFICATION USING MULTISPECTRAL IMAGES

A PROJECT REPORT

for

SOFT COMPUTING(SWE1011)

in

M.Tech (Software Engineering)

by

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6TH SEM,2023

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Associate Professor,SITE



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Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

School of Information Technology and Engineering

April,2023

DECLARATION BY THE CANDIDATE

We here by declare that the project report entitled “**Land use Land Cover Classification using Multispectral images**” submitted by us to Vellore Institute of Technology University, Vellore in partial fulfillment of the requirement for the award of the course **Soft Computing (SWE1011)** is a record of bonafide project work carried out by us under the guidance of **Prof. Agilandeewari L.** We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other course.

Place : Vellore

Signature:

Date :



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School of Information Technology and Engineering[SITE]

CERTIFICATE

This is to certify that the project report entitled “**Land Use Land Cover Classification using Multispectral Images**” submitted by **YENIREDDY JAHNAVI(20MIS0161),KANEKANTI.TEJASREE(20MIS0265),LAKSHMI G MENON(20MIS0346)**to Vellore Institute of Technology University, Vellore in partial fulfillment of the requirement for the award of the course **Soft Computing (SWE1011)** is a record of bonafide work carried out by them under my guidance.

Prof. Agilandeewari L

GUIDE

Asso. Professor, SITE

ABSTRACT:

Land use and land cover (LULC) classification are one of the important tasks to monitor the various land cover resources of the earth. Outpacing the other machine learning techniques in classifying images, we have witnessed a growing interest of the remote sensing community in employing these techniques for the land use and land cover classification based on multispectral and hyperspectral images. This research study uses Remote Sensing (RS) to detect and quantify LULC changes that occurred in the city throughout a twenty-year study period, using Landsat 7 Enhanced Thematic Mapper (ETM+) images from a range of twenty years and Landsat 8 Operational Land Imager (OLI) images from 2019. A supervised classification method using an Artificial Neural Network (ANN) was used to classify and map LULC types. The kappa coefficient and the overall accuracy were used to ascertain the classification accuracy. Therefore, considering the LULC classification performance and data characteristics, GF-2 satellite data could serve as a valuable and reliable high-resolution data source for land surface monitoring. Future works should focus on improving LULC classification accuracy by exploring more classification features and exploring the potential applications of GF-2 data in related Applications.

INTRODUCTION:

In this study, we address the challenge of land use and land cover classification using Sentinel-2 satellite images. The Sentinel-2 satellite images are openly and freely accessible provided in the Earth observation program Copernicus. We present a novel dataset based on Sentinel-2 satellite images covering 13 spectral bands and consisting out of 10 classes with in total 27,000 labeled and geo-referenced images. We provide benchmarks for this novel dataset with its

spectral bands using state-of-the-art deep Convolutional Neural Network (CNNs). With the proposed novel dataset, we achieved an overall classification accuracy of 98.57%. The resulting classification system opens a gate towards a number of Earth observation applications. We demonstrate how this classification system can be used for detecting land use and land cover changes and how it can assist in improving geographical maps.

Problem Statement:

Human activities are one of the main driving forces of land use and land cover changes, and generally in natural environments. Land-use change affects the nature of the climate, cause natural hazards and socio-economic dynamics on a global and local scale. Therefore, defining the land-use change is an important issue considering global dynamics and their response to the water flow, forest, soil, and rainfall and runoff characteristics of the catchment or basins. In addition to that investigating land use, land cover change is important for effective sustainable land resource management, effective utilization of the natural resource, and use as a tool for policymakers and understanding the changes in hydrological processes.

Dataset Description:

EuroSAT is a land use and land cover classification dataset. The dataset is based on Sentinel-2 satellite imagery covering 13 spectral bands and consists of 10 LULC classes with a total of 27,000 labeled and geo-referenced images.

Dataset Link:

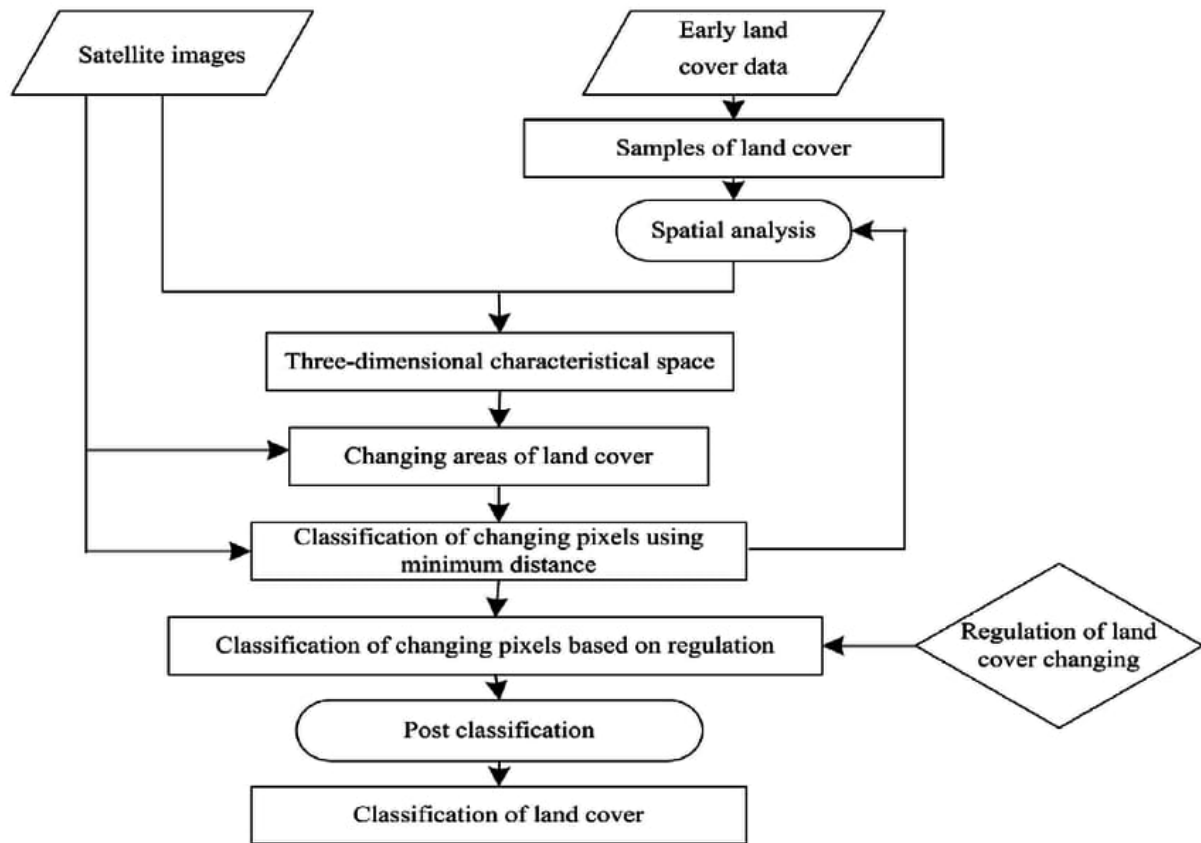
<https://drive.google.com/drive/folders/1FCzyPMO3T4sY62FjVU6-moNoOTF1kMNH>

Methodologies used:

Mapping and identifying land cover/land use and its change is the most important, as well as the most widely researched, topic in remote sensing. Land cover/land use has been used extensively to derive a number of biophysical variables, such as vegetation index, biomass, and carbon content . More importantly, land cover/land use pattern and its change reflect the underlying natural and/or social processes, thus providing essential information for modeling and understanding many different phenomena on the Earth.

Knowledge of land cover/land use and its change is also critical to effective planning and management of natural resources. Mapping land cover/land use accurately and efficiently via remote sensing requires good image classification methods. Unfortunately, there are numerous factors that could affect the effectiveness and accuracy of the classification algorithms. Different land cover/land use classification methods may be needed for different problems under different environmental conditions, making generalization and hence automation of the image classification process across time and space extremely difficult.

Architecture Diagram :



Literature Survey:

AUTHORS	METHODOLOGIES OR TECHNIQUES USED	ADVANTAGES	ISSUES	METRICS USED
H.A.Kaul	Surface irrigation Well irrigation Three major irrigation projects.	1.Classifying and mapping is an integral step in understanding the	The images had to be Normalized and subset	Accuracy of classification methodology

	11 Medium irrigation projects, 62 minor irrigation projects and 315 percolation tanks are completed.	earths bio-physical systems. 2. Different land cover surfaces reflect sunlight in distant wave lengths of electromagnetic spectrum.	for input to the classification and detection process.	is 89% and 91.02%.
Nirav Desai, Dr.Parag Shukla	Imperial technique, Remote sensing methodology.	1.It is cost effective. 2.It consumes lesser amount of time.	To comprehend the influence of anthropological activities on or earth.	SVM has the better classification accuracy.
L.E.Christovam,Guilherme Ghomes pessoa,Milton Hirokazu,Shimabukuro,Maria De lourdesBueno Trindade Galo	Spectral angle mapper(SAM), Support vector machine(SVM), and Random forest.	Adding more features rather than increasing the classification accuracy may actually decrease the accuracy.	Accuracy of remote sensing image classification defining the image and classification algorithm.	Accuracy.
Ava vali , Sara comai , Matteo Matteucci	Machine learning and deep learning techniques	1.Deep learning has been successful in dealing with big data. 2.Deep learning has an enormous capacity to act as an indispensable tool to tackle most serious	Limited resolution, high dimensionality, redundancy within data, atmospheric and acquisition noise, calibration	F – Score

		environmental concerns.	of spectral bands and many other source specific issues.	
Pramod Soni , Navin Rajpal , Rajesh Mehta,Vikas h Kumar Mishra	Kappa co-efficient overall accuracy and accuracy of users and producers.	Human skin utilize this data in agriculture, forestry, disaster management urbanization and many more applications.	The high complex nature of study area makes the classification task challenging and appealing.	All methods have more than 90% Accuracy.
Darshana Rawal , Abha Chhabra , Mehul Pandya , Anjana Vyas	Object based segmentation, NIDVI, pixel based classification	The pre-processing and post image processing are carried out to enhance quality of images.	Many potential users remain sceptical about the capabilities of remote sensing data.	Overall accuracy of map is 84.48% and users accuracy as 89.26% and 84.47%.
Loukika K N , Venkata Reddy Kesara , Venkataramana Sridhar	Google earth engine, landsat -8, sentinell-2, cloud masking, NDVI, Accuracy assessment.	Accuracy assessment of each individual class can be used to evaluate the performance of each classifier with respect to each class.	1.SVM is sensitive to hyper parameters. 2.RF is more resilient and less impacted by parameters.	Accuracy.
Saeid Amini , Mohsen Saber, Hamidreza Rabiei-Dastjerdi ,	Statistical pan sharpening algorithm, Lulc mapping , Fuzzy classification , SVM, Random forest, Deep learning,DTL.	1.Reliable land use classification. 2.Continuous increase in urbanization characterises	Big data analytic in RS.	F-Score

Saeid Homayouni		countries in middle east.		
Kun Jia , Jingcan Liu , Yixuan Tu , Qiangzi Li	Pixel based and object based classifications, multi resolution segmentation method,SVM,RBF.	Assessment of LULC classification performance using multi-spectral data have significant work.	Aquiring a representative sample collection is the most time consuming process.	Users accuracy of 85.48% and producers accuracy of 93.43%.
Hayder Dibs , Hashim Ali Hasab , Jawad K . AI – Rifaie, Nadhar AI - Ansari	Landsat-8 multispectrum, image re-sampling, radio metric and geometric collection images layer stacking.	Different classification methods were adopted to classify the images MD and MLC classifiers.	Effectiveness of using panchromatic have highly spatial resolution to refine the methodology.	Results having overall accuracy and kappa co-efficient of 97.78% and 0.93%.
Jane Ferah Gondwe	Landsat imagery,Image Pre processing , Selection of training data , post classification .	Most of the herbaceous land was converted to agricultural land to serve the need for food for the growing population.	The expansion of built up land was at the expense of forest land ,agricultural land.	Accuracy
T.Vignesh , R. Beulah Jeyavathana , K.K.Thyagarajan , K.V.Kanimozhi	Recurrent neural network	1.It provides higher accuracy than any other methods. 2.RNN is able to classify multi-spectral images into various land cover classes.	For small areas assessing land cover classes is easy but for remote areas it is very difficult.	Accuracy

Jiahui Xu , Weimin Ju , Zhongwen Hu	Multi scale marker based water shed segmentation, classifier confusion.	Improves the land cover classification accuracy effectively.	Remote sensing images are often quite complex and some classifiers may perform better in terms of accuracies for some component s of data.	Accuracy.
Daniela I Moody , Steven P.Brumby , Joel C.Rowland	Clustering on sparse approximations, architect satellite imagery, dictionary learning algorithm, multi-spectral under complete learned dictionaries.	In the case of mixed classifications are method could be more efficient than trying to classify one class at a time using a subjective supervised method.	Making the cosa method successful for the classificatio n in remote sensing will be to preconditio n the training set and the learning algorithm.	Accuracy.
Miae Kim, Junghee Lee, Daehyeon Han , Minso Shin	Generation of 2-D spectral curve images, Convolutional neural networks, RF and SVM's, Accuracy assesment	Deep learning techniques such as convolutional neural network that overcome the well known limitations of ANN's have recently gained popularity for application to land cover classification.	Urban areas are a mixture of land cover types such as soil and vegetation as well as imperial surfcaes.	Accuracy

Khyat Patel , Manan Jain , Manish I . Patel , Ruchi Gajjar	Extracting RGB pixels, generating pixel data set, extracting ROI features, SVM, RFC, NN.	1.The reasearch work is able to perform the classification of land covers for the target areas. 2.It have Less expensive ML techniques which provides comparable accuracy with more complex techniques.	With the rapid change in land use patterns,the land covers are varying drastically which requires immediate attention.	SVM classifier achieved the highest Accuracy of 98.48%.
Peijun Li , Xiaobai Xiao	Water shed segmentation	1.Image classification is a straight forward method. 2.Image classification for land cover mapping is performed on the image using maximum likelihood method.	If more than 50% of the pixels of particular segment are categorized in one class then the segment was considered to be both existing and labelling, if not then pixels of the segment are classified according to the 3X3 majority filter pixel.	Precision

Dr . G Stavrakoudis , J.B Theocharis	Boosted Genetic fuzzy classifier	Both the stages of BGFC's learning algorithm uses evolution strategies as a search algorithm	High interpretability. The required cooperation among the rules is implicitly induced during their creation	Precision
Anand Upadhyay , Santosh Kumar singh , Aditya Shetty , Zibreel siddiqui	Pre processing and feature extraction , Design of decision tree and KNN classifiers , Training of classification algorithm , Testing of classifiers.	1.Land use land cover plays important role in sustainable development 2.Land use land cover helps to understand the and plan for the sustainable development.	In an unsupervised classification there is no involvement of the training data sets and there is nothing as such learning between feature and classes.	Accuracy
Dinesh sathyanarayanan, D V Anudeep , R Hebbar , K.Ganesha Raj	Data set preprocessing , Modified segnet for multiclass classification , Training configuration ,	Machine learning , fast evolving sub disciplines like DL come with the promise of making analysis of satellite images easier and more scalable and show a scope for full or partial automation of spectral data analysis.	Labeling of training samples,	Precision and F-Score

K.C Tan , H.S.Lim and M.Z. Mat Jafri	Neural network , Maximum Likelihood	Remote sensing offers the advantage of rapid data acquisition of land cover information at lower cost as compare with ground truth data acquisition.	Neural network is increasingl y utilized for the interpretati on of rremotely sensed data due to its capability to accurately approximat e complex non linear functions.	Accuracy
Tuan Linh Giang , Kinh Bac Dang , Quang Toan le , SI son Tong	Random forest, SVM.	The integration can reduce the time and cost to take samples when users want to interrupt land cover types from a new UAV images.	The accuracy of both training and testing data was checked during UNet developme nt	Acuuracy.
Barnaby Clark , Petri Pellikka	Data and software, Image preprocessing , Maximum likelihood technique ,Classification nomenclature	Remote sensing offers the most efficient methodology for routinely monitoring at landscape level over regional as well as national and continental scales.	Nevertheles s mapping the landscape of the Taita hills from this imagery represents a challenge because of the limited spectral information and the	Accuracy

			heterogeneous land cover .	
Wei qi zhou	Data pre-processing, Land cover classification, Classification using LIDAR data alone ,Classification Integrating LIDAR nDSM and imagery.	Lidar is an active remote sensing technology than passive multispectral remote sensing .	Classification is more challenging due to the fine scale complexity of urban land cover features .	F-Score
Veeramallu Satya Sahithi , Stalin Subbiah , Shefali Agarwal	Pre processing,data/image fusion , Image classification	The colour normalized transform is a fusion technique used for fusing two multispectral images that uses a mathematical combination of colour image and a high resolution image	Spectral image is smaller than number of bands that are little.	Accuracy
Md.Omar Sarif , Rajan Dev Gupta	Multispectral imageries of land-sat the Matic mapper and operational land imager or thermal infrared sensor of four different time points, Maximum likelihood classifier.	For extraction of change detection two different time points have been incorporated to carry out the periodic change over the land scale.	The need to initiate immediate efforts for balanced and sustainable development	Accuracy.
Zhang Rongqun , Zhu Daolin	Knowledge – rule method , Gray level co occurrence matrix	1.The four multispectral bands in the quick bird data were transferred into four principal components by PCA.	The spectral and spatial information in the images could be fully extracted and then all	Accuracy

		2.Can effectively increase the classification accuracy of ground objects and could provide theoretical bases and approaches for autoimmunization of land cover.	of the information could be comprehensively applied.	
Fusun Balik Sanli , Saygin Abdikan , Mustafa Tolga Esetlili , Filiz sunar	Image pre-processing application of fusion techniques, quality analysis of fused images, land use of cover mapping and accuracy assessment of classification results.	HPF method allows to use more than 3 bands, Ehlers methods has developed to keep the spectral characteristics during image fusion.	It is uncertain that the preference of c- band or l-band for land use land cover analysis, dry soil and moisture soil areas differ prominently due to differences in back scatter values.	Accuracy.
Wen ting Cai , Yong xue Liu , Man chun Li , Yu Zhnag , Zhen Li	Best first method, Best first DT, Standard DT	Using Decision trees to classify remotely sensed data has increased due to algorithm running fast and making statistical assumptions.	Calculating features for classification, Determining the best feature space dimension , avoiding overfitting the training data.	F-Score

P.das, V.Pandey	Binary logistic technique, moderate resolution, sentinel-2 multi-spectral data, BLR, Random forest.	Simple working principle, Quick data processing and high accuracy are providing great advantages over other parametric and non-parametric complex machine learning approaches and offering potential scopes for further improvement.	Low producer accuracy was observed for barren land.	R-Software.
Mansour Halimi , Zahra Sedighifar , Chenour Mohammadi	Satellite data acquisition, image visual interpretation, maximum likelihood classification, accuracy assessment.	Satellite remote sensing coupled with GIS, NDA powerful tool for mapping and evaluation of land use, land cover.	Lack of valuation of ecological services, Ignorance of bio physical limitations and use of ecological incompatible technologies	Precision.
Victor Alhassan, Christopher Henry, Sheela Ramana, Christopher Storie	Deep learning methods, FCN-VGG,	FCN -Google linnet liberally employs max pooling operations within its inception modules and across its structure. 2. The seed in training FCN – google net is very noticeable	Process of LULC mapping is achieved using deep neural networks pre trained on the image net large scale maps.	Accuracy

		compared to the other base networks ,but performance gains quickly plateaus after a number of epochs.		
Riad Arefin , Sarita Gajbhiye Meshram , Celso Augusto Guimaraes Santos	Image processing, multispectral and panchromatic image fusion.	Carry out analysis using the maximum likelihood classification on multi spectral data by means of qualitative and quantitative approaches.	During the classification the vegetation, built-up and cultivate land classes are biased and slightly deviated from the accuracy	Accuracy
Patrick Helber , Benjamin Bischke , Andreas Dengel , Damian Borth	Image classification methods, Land use land cover classification.	The data sets are based on very high-resolution images with a spatial resolution of up to 30cm per pixel.	1.The sentinel-2 satellite images are openly and freely accessible provided in earth observation program. 2.Classification of remotely sensed images.	Accuracy.
Ava vali, Sara Comai, Matteo Matteucci	Hyper spectral image classification, pixel level classification.	The fast-growing volume of timely data available at the global scale offer, new opportunities for	Strongly impacts the performance of classification.	Accuracy.

		a variety of applications.	2.To deal with the data that comes with the variety of impacting variables.	
Xu, Bing Gong, PengA	Principal component analysis, segmented PCA, segmented IDA, penalized discriminant analysis.	After level 1 classification the HS image generated thematic map that is greater in kappa co-efficient comparing to the MS image.	Classification shows that HS data does not produce better results than Ms data.	Accuracy.
J.Jayanth , V.S Shalini, T.Ashok Kumar , Shivaprakash Koliwad	Elephant herding algorithm.	When the spectral resolution of satellite imagery is increased the higher within class variability reduces the statistical separability between land use land cover classes	Experimental result has revealed that elephant herding algorithm shows an improvement of 10.7%.	Accuracy.
Md.Sazzad Hossain, Md Asif Haider Khan, Tomiwa Victor oluwajuwon, Jayanta Biswas	FWKS method.	LULC assessment is challenging for vast forest ecosystem where satellite images allow for forecasting.	The comparative potential of vegetation indices including normalized difference vegetation index and soil adjusted vegetation index in	Accuracy.

			forest cover assessment.	
Neha Kumari, Sonajharia Minz	SVM classifiers, DRSVM.	The DRSVM demonstrator higher efficiency in terms of computational parameters and time complexity.	Conventional supervised deep learning algorithms like CNN over fit in dense layers in case of inadequate training data.	Accuracy.
Akshar Tripathi, Shashi Kumar, Sandeep Maithani	Remote sensing,	1.Arbitrary number of image bands can be used. 2. Low resolution multi spectral images form the uncorrelated principal components	Deep learning model is unavailability of labelled training samples. Creating a label data set is a cumbersome and time consuming task.	Accuracy
Taposh Mollick, Md Golam Azam, Sabrina Karim	Pixel based and object-based image classification, ERDAS imagine tools.	LULC classification using RS techniques such as use of satellite can forecast the crop yield and provides information on weeds.	It frequently generates a lot of discontinuity in a particular land cover resulting in small objects	Accuracy.

			within land cover	
Chunyu Li, Rong Cai, Wei tian , Junna Yuan,Xiaofe i Mi	Data pre-processing, CART method, feature extraction, evaluation metrics.	High temporal resolution, high spatial resolution, various sensors are deployed on the gaofem series satellites.	Remote sensing data is accurate fast and intelligentl y gathered and the information coming from remote sensing satellite big data can be mined.	Accuracy.
Wuttichai, Boonpook, Yumin Tan, Attawut Nardkulpat Kritanai Torsi, Peerapong Torteeka, Montri Jainaen	Land use data set, Accuracy assessment methods, Accuracy assessment algorithms.	The multi- spectral bands shows the differences in spectral reflectance and land use features.	Provides more accurate segmentati on results in agriculture and forest classes.	Accuracy.
Umut Gunes Serfrick, Taskin Kavzoglu Ismail	Random Forest, Extreme gradient boosting algorithms, Ground control points.	High resolution, low cost, rapidly and periodically achievable, 3D data.	UAV's have been improved by integrating real-time kinematic positioning and multi- spectral image equipment.	Accuracy.

Yanchen wu, yinghai ke, huili gong, Beibei chen, linzhu.	Object based classification, based on spectral, texture and geometric features, pixel based classification based on texture features only in panchromatic image, SVM, pixel based SVM.	Pixel based method is advantageous over object based method because of its higher accuracy.	Results show that object based method obtained higher overall accuracy than pixel based method with SRS 1 but lower accuracy with SRS 2.	Accuracy.
Giorgio licciardi, chiara pratola, Fabio del frate.	Land cover neural network, semi- automatic supervised data mining methods.	The selection of suitable training set that represents all the different types of surfaces is crucial for the success of classification process.	If the returned probability is lower than certain threshold the pixel is not assigned to any class, number of classes vary from image to image also the internal layers of NN vary in an automatic way depending on number of outputs.	Precision.

Arijit laha, Nihkhil, R.Pal, fellow, jyotirmoy das.	Different fuzzy methodologies, aggregation methods.	Sofm can avoid the under utilization of prototypes.	Classificati on scheme using contextual information can be more efficient than non- contextual one.	F-Score.
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Code and Implementation:

The screenshot shows a Jupyter Notebook interface with the title 'LULC classification code'. The notebook is running on a local host. The code in the first cell is as follows:

```
In [9]: # Essential and common packages
import os
import glob

# Read and visualize the raster data
import rasterio
from rasterio.plot import show

# Plots and bars
import matplotlib.pyplot as plt
import seaborn as sns

# Computation library
import numpy as np

# Tensorflow for building the resnet50 model
import tensorflow.python.keras as k
import tensorflow as tf
from tensorflow.keras.layers import Input, Add, Dense, Activation, ZeroPadding2D, BatchNormalization, Flatten, Conv2D, AveragePooling2D
from tensorflow.keras.initializers import random_uniform, glorot_uniform
from tensorflow.keras.models import Model

# Sklearn for confusion matrix
import itertools
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

# For visualization of plots without plt.show()
%matplotlib inline
```

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For visualization of plots without plt.show()
%matplotlib inline

Define the required variable ¶

```
In [2]: dataset_url = r'C:\Users\tek\Desktop\try\DL-for-LULC-prediction\EuroSAT\2750'  
batch_size = 32  
img_height = 64  
img_width = 64  
validation_split=0.2  
rescale=1.0/255
```

Data preparation for the model

```
In [3]: datagen = tf.keras.preprocessing.image.ImageDataGenerator(validation_split=validation_split, rescale=rescale)  
dataset = tf.keras.preprocessing.image_dataset_from_directory(dataset_url, image_size=(img_height, img_width),
```

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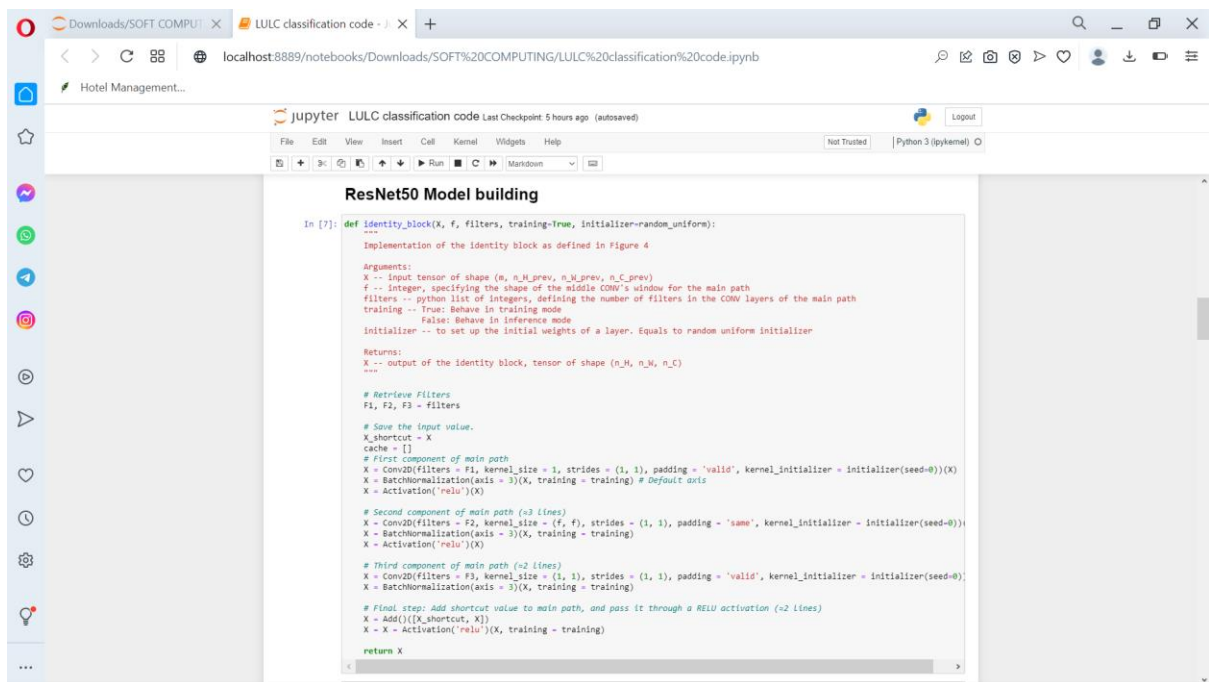
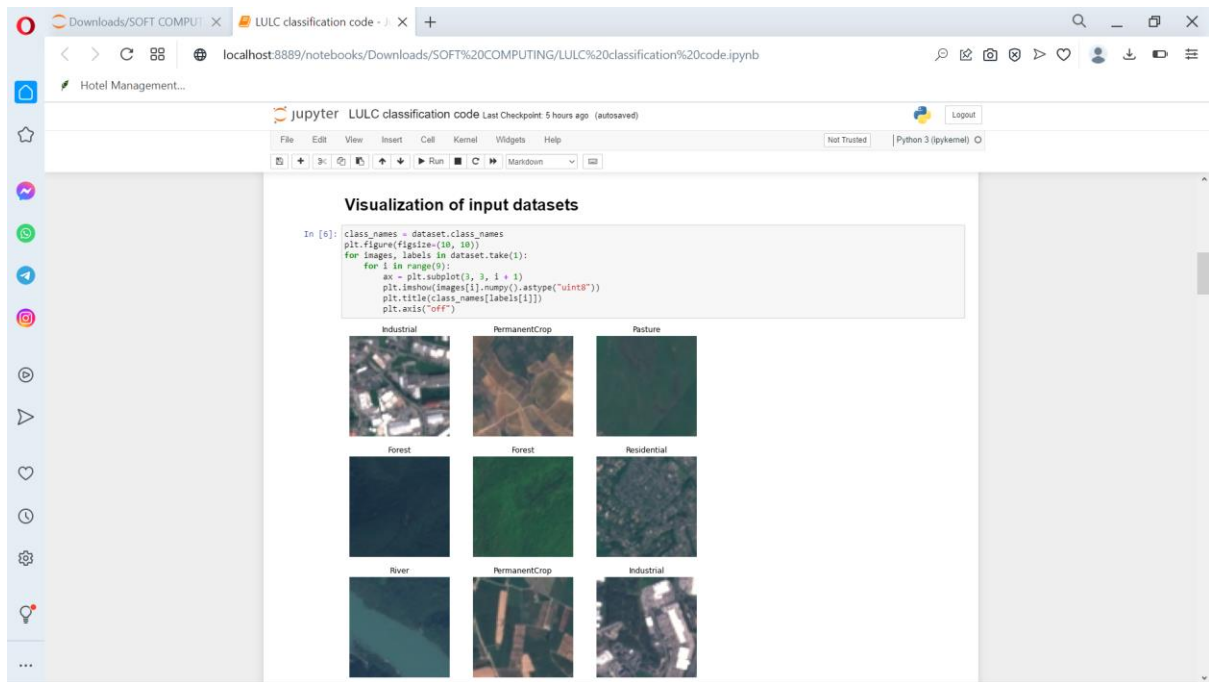
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Data preparation for the model

```
In [3]: datagen = tf.keras.preprocessing.image.ImageDataGenerator(validation_split=validation_split, rescale=rescale)  
dataset = tf.keras.preprocessing.image_dataset_from_directory(dataset_url, image_size=(img_height, img_width), batch_size=batch_size)  
  
Found 27000 files belonging to 10 classes.
```

```
In [4]: train_dataset = datagen.flow_from_directory(batch_size=batch_size,  
directory=dataset_url,  
shuffle=True,  
target_size=(img_height, img_width),  
subset="training",  
class_mode='categorical')  
  
Found 21600 images belonging to 10 classes.
```

```
In [5]: test_dataset = datagen.flow_from_directory(batch_size=batch_size,  
directory=dataset_url,  
shuffle=True,  
target_size=(img_height, img_width),  
subset="validation",  
class_mode='categorical')  
  
Found 5400 images belonging to 10 classes.
```



```
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In [8]: def convolutional_block(X, f, filters, s = 2, training=True, initializer=glorot_uniform):
    """
    Implementation of the convolutional block as defined in Figure 4

    Arguments:
    X -- input tensor of shape (m, n_H_prev, n_W_prev, n_C_prev)
    f -- integer, specifying the shape of the middle CONV's window for the main path
    filters -- python list of integers, defining the number of filters in the CONV layers of the main path
    s -- Integer, specifying the stride to be used
    training -- True: Behave in training mode
    False: Behave in inference mode
    initializer -- to set up the initial weights of a layer. Equals to glorot uniform initializer,
    also called Xavier uniform initializer.

    Returns:
    X -- output of the convolutional block, tensor of shape (n_H, n_W, n_C)
    """

    # Retrieve Filters
    F1, F2, F3 = filters

    # Save the input value
    X_shortcut = X

    ##### MAIN PATH #####
    # First component of main path glorot_uniform(seed=0)
    X = Conv2D(filters = F1, kernel_size = f, strides = (s, s), padding='valid', kernel_initializer = initializer(seed=0))(X)
    X = BatchNormalization(axis = 3)(X, training=training)
    X = Activation('relu')(X)

    # Second component of main path (=2 lines)
    X = Conv2D(F2, (f, f), strides = (1, 1), padding = 'same', kernel_initializer = initializer(seed=0))(X)
    X = BatchNormalization(axis = 3)(X, training = training)
    X = Activation('relu')(X)

    # Third component of main path (=2 lines)
    X = Conv2D(F3, (1, 1), strides = (1, 1), padding = 'valid', kernel_initializer = initializer(seed=0))(X)
    X = BatchNormalization(axis = 3)(X, training = training)

    ##### SHORTCUT PATH ##### (=2 lines)
    X_shortcut = Conv2D(F3, (1, 1), strides = (s, s), padding = 'valid', kernel_initializer = initializer(seed=0))(X_shortcut)
    X_shortcut = BatchNormalization(axis = 3)(X_shortcut, training = training)

    # Final step: Add shortcut value to main path (use this order [X, X_shortcut]), and pass it through a RELU activation
    X = Add([X, X_shortcut])
    X = Activation('relu')(X)
    return X
```

```
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# Third component of main path (=2 lines)
X = Conv2D(F3, (1, 1), strides = (1, 1), padding = 'valid', kernel_initializer = initializer(seed=0))(X)
X = BatchNormalization(axis = 3)(X, training = training)

##### SHORTCUT PATH ##### (=2 lines)
X_shortcut = Conv2D(F3, (1, 1), strides = (s, s), padding = 'valid', kernel_initializer = initializer(seed=0))(X_shortcut)
X_shortcut = BatchNormalization(axis = 3)(X_shortcut, training = training)

# Final step: Add shortcut value to main path (use this order [X, X_shortcut]), and pass it through a RELU activation
X = Add([X, X_shortcut])
X = Activation('relu')(X)
return X

In [9]: def ResNet50(input_shape = (64, 64, 3), classes = 6):
    """
    Stage-wise Implementation of the architecture of the popular ResNet50:
    CONV2D -> BATCHNORM -> RELU -> MAXPOOL -> CONV2D -> CONV2D -> IDBLOCK*2 -> CONV2D -> IDBLOCK*3
    -> CONV2D -> IDBLOCK*5 -> CONV2D -> IDBLOCK*2 -> AVGPOL -> FLATTEN -> DENSE

    Arguments:
    input_shape -- shape of the images of the dataset
    classes -- Integer, number of classes

    Returns:
    model -- a Model() instance in Keras
    """

    # Define the input as a tensor with shape input_shape
    X_input = Input(input_shape)

    # Zero-padding
    X = ZeroPadding2D((3, 3))(X_input)

    # Stage 1
    X = Conv2D(64, (7, 7), strides = (2, 2), kernel_initializer = glorot_uniform(seed=0))(X)
    X = BatchNormalization(axis = 3)(X)
    X = Activation('relu')(X)
    X = MaxPooling2D((3, 3), strides=(2, 2))(X)

    # Stage 2
    X = convolutional_block(X, f = 3, filters = [64, 64, 256], s = 1)
    X = Identity_block(X, 3, [64, 64, 256])
    X = Identity_block(X, 3, [64, 64, 256])

    # Stage 3 (=4 lines)
```

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```
# Stage 2
X = convolutional_block(X, f = 3, filters = [64, 64, 256], s = 1)
X = identity_block(X, 3, [64, 64, 256])
X = identity_block(X, 3, [64, 64, 256])

# Stage 3 (4 lines)
X = convolutional_block(X, f = 3, filters = [128, 128, 512], s = 2)
X = identity_block(X, 3, [128, 128, 512])
X = identity_block(X, 3, [128, 128, 512])
X = identity_block(X, 3, [128, 128, 512])

# Stage 4 (4 lines)
X = convolutional_block(X, f = 3, filters = [256, 256, 1024], s = 2)
X = identity_block(X, 3, [256, 256, 1024])
X = identity_block(X, 3, [256, 256, 1024])
X = identity_block(X, 3, [256, 256, 1024])
X = identity_block(X, 3, [256, 256, 1024])

# Stage 5 (3 lines)
X = convolutional_block(X, f = 3, filters = [512, 512, 2048], s = 2)
X = identity_block(X, 3, [512, 512, 2048])
X = identity_block(X, 3, [512, 512, 2048])

# AVGPOOL (1 line). Use "X = AveragePooling2D(...)(X)"
X = AveragePooling2D(pool_size = (2, 2), name = 'avg_pool')(X)

# output layer
X = Flatten()(X)
X = Dense(classes, activation='softmax', kernel_initializer = glorot_uniform(seed=0))(X)

# Create model
model = Model(inputs = X_input, outputs = X)

return model
```

Model train

```
In [10]: model = ResNet50(input_shape=(64,64,3), classes=10)

In [11]: model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
```

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Model train

```
In [10]: model = ResNet50(input_shape=(64,64,3), classes=10)

In [11]: model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

In [12]: # please increase the epoch for higher accuracy (epochs=100)
history = model.fit(train_dataset, validation_data=test_dataset, epochs=20, batch_size=32)

6/5/6/5 [=====] - 260s 376ms/step - loss: 1.5560 - accuracy: 0.5439 - val_loss: 1.1880 - val_accuracy: 0.6430
Epoch 2/20
6/5/6/5 [=====] - 254s 376ms/step - loss: 1.0657 - accuracy: 0.6463 - val_loss: 1.2341 - val_accuracy: 0.6067
Epoch 3/20
6/5/6/5 [=====] - 255s 378ms/step - loss: 0.9766 - accuracy: 0.6866 - val_loss: 1.0293 - val_accuracy: 0.6533
Epoch 4/20
6/5/6/5 [=====] - 255s 377ms/step - loss: 0.8460 - accuracy: 0.7325 - val_loss: 0.7216 - val_accuracy: 0.7031
Epoch 5/20
6/5/6/5 [=====] - 254s 376ms/step - loss: 0.7362 - accuracy: 0.7642 - val_loss: 0.8269 - val_accuracy: 0.7193
Epoch 6/20
6/5/6/5 [=====] - 253s 375ms/step - loss: 0.6699 - accuracy: 0.7824 - val_loss: 0.6173 - val_accuracy: 0.7919
Epoch 7/20
6/5/6/5 [=====] - 254s 376ms/step - loss: 0.5544 - accuracy: 0.8200 - val_loss: 0.5773 - val_accuracy: 0.8122

In [13]: model.save('lulc_20_epoch')
```

C:\Users\tek\anaconda3\envs\tf\lib\site-packages\tensorflow\python\keras\utils\generic_utils.py:494: CustomMaskWarning: Custom mask layers require a config and must override get_config. When loading, the custom mask layer must be passed to the custom_objects argument.
warnings.warn('Custom mask layers require a config and must override ')

INFO:tensorflow:Assets written to: lulc_20_epoch/assets

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Run

```
C:\Users\tek\anaconda3\envs\tf\lib\site-packages\tensorflow\python\keras\utils\generic_utils.py:494: CustomMaskWarning: Custom mask layers require a config and must override get_config. When loading, the custom mask layer must be passed to the custom_objects argument.
warnings.warn('Custom mask layers require a config and must override '
INFO:tensorflow:Assets written to: lulc_20_epoch\assets
```

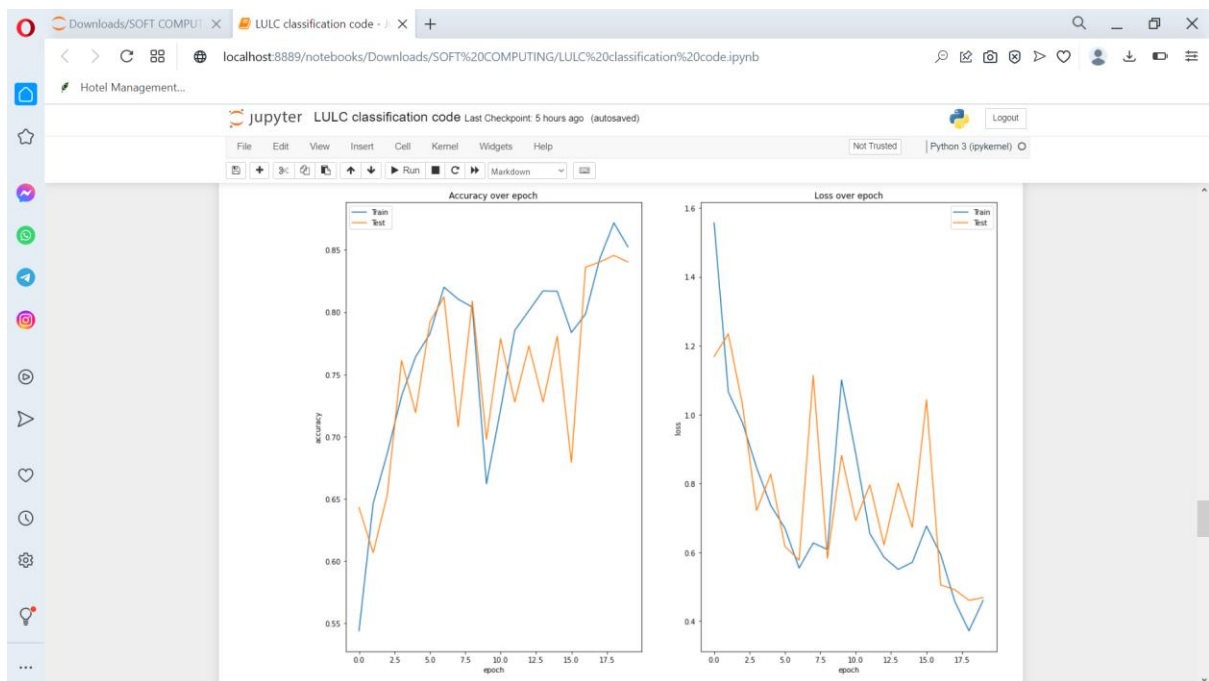
analyzing results and visualization

```
In [14]: fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16.53, 11.69))
ax1.plot(history.history['accuracy'])
ax1.plot(history.history['val_accuracy'])
ax1.set_xlabel('epoch')
ax1.set_ylabel('accuracy')
ax1.set_title('Accuracy over epoch')
ax1.legend(['Train', 'Test'], loc='upper left')

ax2.plot(history.history['loss'])
ax2.plot(history.history['val_loss'])
ax2.set_xlabel('epoch')
ax2.set_ylabel('loss')
ax2.set_title('Loss over epoch')
ax2.legend(['Train', 'Test'], loc='upper right')

Out[14]: <matplotlib.legend.Legend at 0x263ba4c8490>
```

Accuracy over epoch Loss over epoch



```
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Confusion matrix

In [15]: y_pred = [] # store predicted labels
y_true = [] # store true labels

# iterate over the dataset
for i, (image_batch, label_batch) in enumerate(test_dataset): # use dataset.unbatch() with repeat
    # append true labels
    y_true.append(label_batch)
    # compute predictions
    preds = model.predict(image_batch)
    # append predicted labels
    y_pred.append(np.argmax(preds, axis = 1))
    if i==300:
        break

# convert the true and predicted labels into tensors
correct_labels = tf.concat([item for item in y_true], axis = 0)
correct_labels = np.argmax(correct_labels, axis=1)
predicted_labels = tf.concat([item for item in y_pred], axis = 0)

In [16]: cm = confusion_matrix(correct_labels, predicted_labels)
cm

Out[16]: array([[ 873,   4,  11,  29,   5,  31,  62,   0,  16,  22],
 [   5, 972,   4,   1,   0,  44,   0,   0,   9,  31],
 [  26,  36, 780,  28,  25,  15,  95,  36,  28,   8],
 [  18,   1,  16, 695,  49,  16,  34,  33,  40,   2],
 [   0,   0,   5,  19, 828,   0,   2,  43,  17,   0],
 [  23,  12,  14,  26,   0, 585,  25,   0,  12,  11],
 [  40,   0, 103,  48,  18,  22, 642,  16,   3,   0],
 [   0,   0,  21,  14,  33,   0,   7, 988,   0,   0],
 [  26,  12,  10,  73,   9,  48,  17,   0, 700,   6],
 [   8,  10,   0,   0,   0,   4,   0,   0,   1, 1047]],
      dtype=int64)
```

```
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In [17]: def plot_confusion_matrix(cm, classes,
                                normalize=False,
                                title='Confusion matrix',
                                figsize=(10, 10),
                                cmap=plt.cm.Blues):
    """
    This function prints and plots the confusion matrix.
    Normalization can be applied by setting 'normalize=True'.
    """
    plt.figure(figsize=figsize)
    plt.imshow(cm, interpolation='nearest', cmap=cmap)
    plt.title(title)
    plt.colorbar()
    tick_marks = np.arange(len(classes))
    plt.xticks(tick_marks, classes, rotation=45)
    plt.yticks(tick_marks, classes)

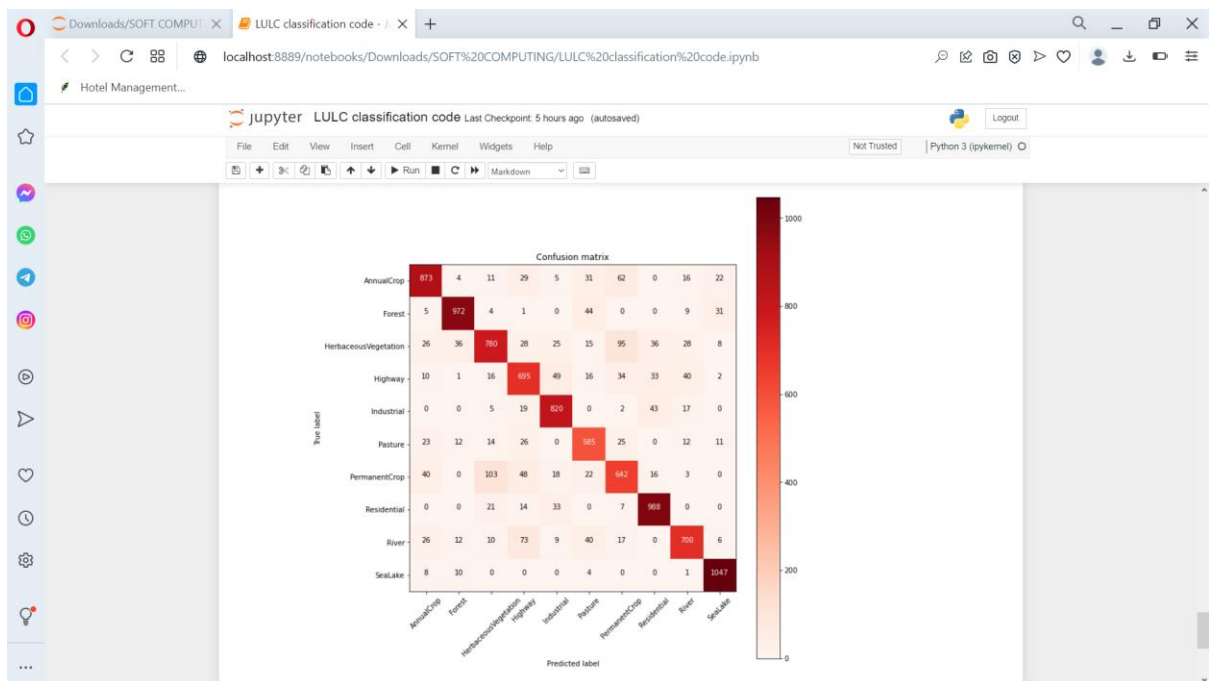
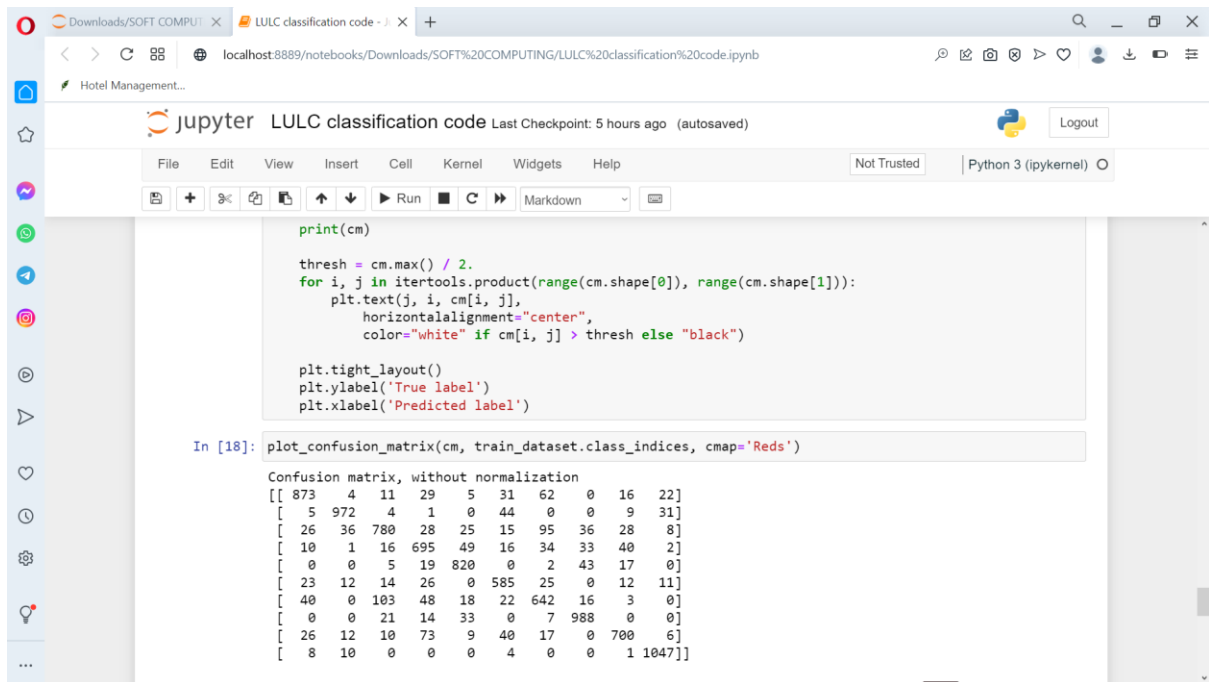
    if normalize:
        cm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
        print("Normalized confusion matrix")
    else:
        print('Confusion matrix, without normalization')

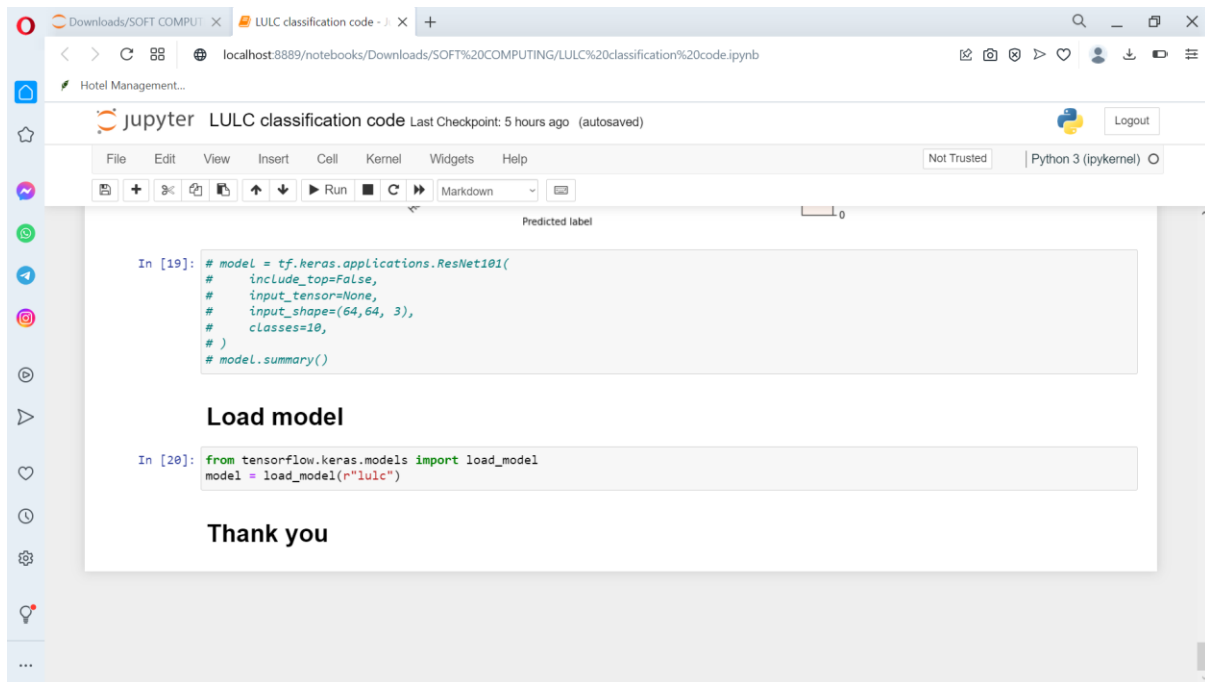
    print(cm)

    thresh = cm.max() / 2.
    for i, j in itertools.product(range(cm.shape[0]), range(cm.shape[1])):
        plt.text(j, i, cm[i, j],
                 horizontalalignment='center',
                 color="white" if cm[i, j] > thresh else "black")

    plt.tight_layout()
    plt.ylabel('True label')
    plt.xlabel('Predicted label')

In [18]: plot_confusion_matrix(cm, train_dataset.class_indices, cmap='Reds')
Confusion matrix, without normalization
```





Conclusion:

The methodology used for the study is capable of generalizing quite well over the entire Jalgaon District, an area of 11,765 sq. km accurately identifying areas of Agriculture, Harvested, Barren, Saline, Forest, Water Body and Settlement. Moreover it was also capable of classifying the different classes in all the 15 major and medium irrigation projected area separately. This study successfully shows that the change detection techniques can be applied to land use land cover change using remote sensing data. Remote Sensing is the most used technology in LULC change detection analysis. The overall accuracy and kappa coefficient values obtained met the classification criteria. The results of the study revealed that there had been a significant change in LULC during the 20-year study period in the area. The obtained results showed that there was a decrease (from the initial area) in the area of forest land, waterbody, bare land, and herbaceousland, while the area of built-up land and agricultural land increased.